Data Structures Laboratory Teaching

Group A: Arrays and Searching Sorting Algorithms

Assignment No 01

Problem Statement: Write a Python program to manage the borrowing records of books in a library. Implement the following functionalities:

1. Compute the average number of books borrowed by all library members
2. Find the book with the highest and lowest number of borrowings in the library
3. Count the number of members who have not borrowed any books(denoted by a borrow count of 0).
4. Display the most frequently borrowed book(i.e., the mode of borrow counts)
5. After performing, determine the time and Space complexity of each operation

Code:

```
# library_borrow_manager.py


# Simple Library Borrowing Record Manager


# Sample data: Member ID and their list of borrowed books
library_data = {
    "M001": ["Book A", "Book B"],
    "M002": ["Book C"],
    "M003": [],
    "M004": ["Book A", "Book A"],
    "M005": [],
    "M006": ["Book B", "Book D"]
}


# 1. Average number of books borrowed
total_members = len(library_data)
total_books_borrowed = sum(len(books) for books in library_data.values())
average = total_books_borrowed / total_members
print("Average books borrowed:", average)
```

```
# 2. Most and least borrowed books
all_books = []
for books in library_data.values():
    all_books.extend(books)

book_counts = {}
for book in all_books:
    if book in book_counts:
        book_counts[book] += 1
    else:
        book_counts[book] = 1

# Most borrowed book
most_borrowed = max(book_counts, key=book_counts.get)
print("Most borrowed book:", most_borrowed)

# Least borrowed book
least_borrowed = min(book_counts, key=book_counts.get)
print("Least borrowed book:", least_borrowed)

# 3. Count members who borrowed 0 books
zero_borrowers = sum(1 for books in library_data.values() if len(books) == 0)
print("Members with 0 borrowings:", zero_borrowers)

# 4. Most frequently borrowed book (mode)
print("Most frequently borrowed book (mode):", most_borrowed)
```

#OUTPUT:

average books borrowed: 1.1666666666666667

Most borrowed book: Book A

Least borrowed book: Book C

Members with 0 borrowings: 2

Most frequently borrowed book (mode): Book A